Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

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Abstract

Rac1 (Ras-related C3 botulinum toxin substrate 1) is a molecular switch that catalyses the hydrolysis of GTP to GDP. Owing to the slow nature of GTP hydrolysis and even slower dissociation of GDP, Rac1 must be assisted by Guanine Exchange Factors (GEFs) and GTPase-activating proteins (GAPs). As a result, GEFs and GAPs control the switching mechanism in Rac1. This regulatory control mechanism, by several GEFs and GAPs, provides a means to modulate the Rac1 activity in several diseases in which it has been implicated. The work described here attempts to understand various dynamical aspects of Rac1 in the presence of GTP, GDP, and guanine exchange factors (GEFs) to locate transient binding sites on the Rac1 surface. We use cosolvents as chemical probes to accelerate pocket-opening due to the induced-fit mechanism. We observe four transient sites on the RAC1 surface. Two of which form the docking site for GEFs. One is observed where the DHR domain containing GEFs from the DOCK family binds. And another is located at the site where the ELMO1 protein interacts with RAC1. In the absence of GEFs, the pocket volumes are too small to be druggable, indicating their transient nature. Using the pocket volume fluctuations, we evaluate their significance for targeting small molecules. Furthermore, we use principal component analysis to understand

the protein dynamics that are essential for the opening and closing of these transient pockets.

Introduction

Small GTPases, a ubiquitous class of proteins, play pivotal roles in various cellular processes. Their functions include signal transduction, vesicle trafficking, cytoskeletal dynamics, and cell division. Ras GTPases regulate cell proliferation, differentiation, and survival. Rho GTPases govern cytoskeletal dynamics and cell migration. Their activity is regulated through a molecular switch mechanism by guanine nucleotide exchange factors (GEFs), GTPase-activating proteins (GAPs), and guanine nucleotide dissociation inhibitors (GDIs).[1] These molecular switches cycle between OFF (GDP-bound inactive) and ON (GTP-bound active) states. This switching mechanism controls various processes like cell growth and differentiation to vesicular and nuclear transport.[1,2]

Small GTPases are proteins that possess conserved domains across several families; made up of six β-sheets surrounded by 5 α-helices. All domains are conserved such that they carry a guanine nucleoside binding site to recognise the guanine base and phosphate groups and the magnesium ion.[1,3] The switch regions regulate the traffic to the guanine nucleoside binding site. It is also well established that GTPases alone cannot hydrolyse GTP effectively. Therefore, GEFs and GAPs play a significant role in assisting the switching mechanism of GTPases. They control the dynamics of the switch regions that facilitate either the hydrolysis of GTP or the dissociation of GDP. GDP:Mg^{2+} dissociates from GTPases very slowly, and therefore, GEFs assist in exchanging GDP for GTP. One of the accepted mechanisms of GEFs in the dissociation of GDP is by electrostatic repulsion of the magnesium ion that forms a strong interaction with GTPases and phosphate groups of GDP. The loss of a magnesium ion weakens the affinity of GDP, which eventually dissociates. Alternatively, some GEFs can rearrange the switch I or II conformations such that GDP's affinity is reduced, facilitating its dissociation. However, GEFs can also combine the two mechanisms to enable GDP dissociation.[4,5]

Furthermore, GAPs stabilise the transition state for GTP:Mg^{2+}:GTPases and facilitate the hydrolysis of the γ-phosphate group from GTP. Taken together, the switching mechanism can be seen as GEFs assist in the GTPase activation, while GAPs assist in GTP hydrolysis. Following GTP hydrolysis, the inactive GTPases prevent further signal transduction.[6] In some classes of GTPases, the active and inactive states are conjugated with GTPases being shuttled between the cell membrane and cytosol. Such GTPases undergo post-translational modifications to carry long-chain fatty acids at their C-terminal to assist bilayer insertion. GDIs[7,8] sequester the lipid component to prevent membrane insertion of GTPases in the cytosol. This coordinated action of GTPases, GEFs, GAPs, and GDIs is essential for various cellular processes. Dysregulation of these components contributes to pathologies like cancer, neurodegeneration, and inflammatory disorders. For instance, mutations in Ras and its regulators are implicated in oncogenesis, highlighting the therapeutic potential of targeting GTPase signalling pathways.[1] Such mutations are known to disrupt the equilibrium between the active and inactive states of GTPases. It has also been observed that some mutations enhance the affinity of GEFs or GAPs that trap the small GTPases either in an activated state or an inactivated state. Small molecule inhibitors targeting GTPase activity or disruptors of GEF/GAP interactions[9] are under development for cancer therapy. However, there are several challenges in targeting GTPases and/or GTPases-GAP/GEF complexes. Firstly, targeting the guanine binding site is challenging owing to the large intracellular concentration of GTP and high affinity.[10] In such cases, finding alternate sites can avoid such competition with endogenous substrates. GTPase hyper-activation or inactivation can be corrected by disrupting the protein-protein interaction (PPI) between GTPases and GEFs/GAPs by targeting the regulatory protein docking sites.

Rac1 plays a crucial role in regulating cellular processes owing to its switching mechanism that recruits several proteins upstream and downstream. PREX1, TIAM1, VAV1 and DOCK2 are Rac1-specific GEFs that regulate many cellular processes and, therefore, are implicated in many Rac-dependent pathologies such as asthma, ... Structurally, PREX1, TIAM1 and VAV1 are DH/PH domain-containing proteins, while DOCK2 contains a DHR domain. Irrespective of their structural diversity, GEFs bind to the switch

I and/or II regions. The nucleotide-binding site is well conserved among the GTPase family. This poses a significant challenge to selectively target Rac1. Furthermore, the molar concentrations of GTPs and picomolar binding affinity make it more difficult for the competing small molecules. The switching mechanism offers a conformational landscape that must be studied in depth to locate other binding sites on the Rac1 surface. The association of the GEFs show the existence of alternative binding sites that might transiently open to allow their docking. Therefore, we embarked to understand the complete dynamical landscape of Rac1 protein bound to either nucleotides (GDP or GTP) or GEFs (PREX1, TIAM1, VAV1 or DOCK2).

In this study, we determined the evolution of RAC1 binding pockets in its nucleotide-bound conformation and in interaction with its partners (holo). We measured using molecular dynamics simulations of the pocket, determining that the nucleotide anchor was constrained by the nucleotide binding, how the nucleotide binding propagates on Rac1, and how GEF binding to Rac1 influences these propagations. Our work allows us to deepen our understanding of the intrinsic mechanical organisation of Rac1, and also revealed the presence of cryptic pockets pre-existing nucleotide and/or GEF binding, as presented after.

Results and discussion

Protein-protein interactions are central to biology by coordinating an intricate network of proteins that are recruited and activated in a pathway to pass the signal to the nucleus to control the rate and amount of protein expression. Such networks are tightly regulated by molecular switches, such as Rac1, which are activated in response to GTP binding and turned off upon GTP hydrolysis. The guanine exchange for such molecular switches is assisted by guanine exchange factors that are recruited depending on the pathway. The exchange factors serve a dual role of nucleotide exchange and recruiting proteins that will pass on the signal further in the network. This work describes our attempt to understand the Rac1 dynamics in different bound and free states. Further, the RAC1-GEF interactions open up an alternative site, and we investigate whether such a site is amenable to small molecule targeting to either prevent GEF binding or act as molecular glues to reduce the dissociation rate of Rac1-GEF complexes. Such a strategy will reduce the rate of switching from inactive to active conformations, and thus reduce the downstream signalling to prevent Rac1-dependent pathologies.

GEF docking sites open transient pockets on the Rac1 surface

We employed the POVME2 tool to identify and measure the pocket volume changes associated with each pocket. To locate new pockets, we simulated Rac1 under different conditions (See **Table 1**). Before identifying the binding pockets, all binding partners, including nucleotides, were deleted from the trajectory. This allowed us to focus entirely on the conformations of Rac1 to identify new pockets. The pockets were identified using the starting conformation of Rac1 X-ray structure obtained under different conditions (See **Table 1**). The pocket volume changes were monitored throughout the course of the trajectory. To understand the volume changes of the nucleotide binding site in response to various Rac1 binding partners(See **Table 1**), we measured and compared the volume changes understand different binding conditions. The pocket volume fluctuations are restricted under 3000Å^3 in the presence of GTP and GDP along with the coordinated magnesium ion (**Figure 1B**). This shows the closing of the nucleotide pocket to initiate the GTP hydrolysis (GTP-bound) or a state that would be likely after the GTP hydrolysis (GDP). We also observed that switch region dynamics was reduced significantly, indicating a stabilising effect of with Mg^{2+} that interacts with the switch regions (**Figure 4D and 4E**). The switch region conformation is also found to be reduced by GEF binding (**Figure 4D and 4E**), suggesting their role in controlling the nucleotide traffic by modulating the switching regions. We tested this hypothesis by simulating a nucleotide-free and GEF-free form of Rac1 (delGDP and delGTP in **Figure 1B**) and found that the mean pocket volume and fluctuations showed a greater tendency to increase, however, only marginally, as maximum volume was still under 4000Å^3. We speculate that the pocket could have opened even more had the simulation time been longer.

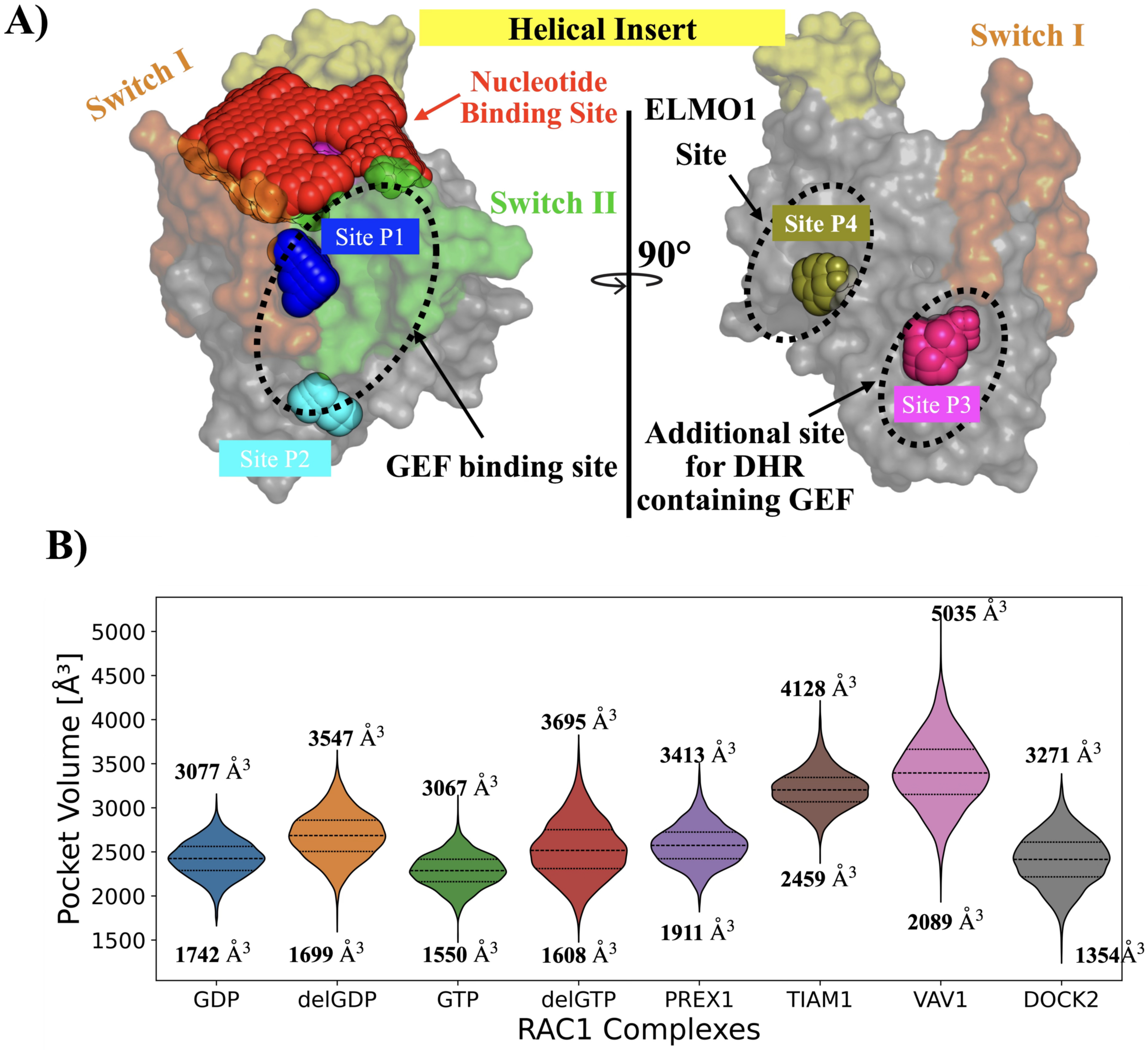

Figure 1: **A**) Rac1 surface with all possible binding pockets. The yellow-coloured region depicts the helical insert, which is peculiar to the Rho GTPase family. The orange-colored region depicts switch I, and the green colored region is switch II. Red spheres occupy the nucleotide-binding site. The pink-colored region is the P-loop. Blue spheres depict Site P1, cyan spheres show Site P2. Magenta spheres occupy the Site P3, and asparagus spheres show the Site P4. **B**) The violin plot illustrates the distribution of the pocket volumes for each of the nucleotide-binding sites. The changes in the pocket volumes are monitored when Rac1 is complexed with either GEFs, GDP, GTP or in the *apo*-states named as delGDP (GDP deleted from GDP-bound system), and delGTP (GDP deleted from GTP-bound system).

Next, we measured the pocket volume and the fluctuations in the presence of GEF in the nucleotide-free state. The Volume fluctuations are higher than in the nucleotide-free state without GEFs. However, the volume changes showed significant changes with different GEFs. For example, PREX1 and DOCK2 showed volume changes that are similar in magnitude to nucleotide-free state without GEFs (See delGDP and delGTP in **Figure 1B**). TIAM1 and VAV1 showed significantly enlarged nucleotide pockets with mean volumes above 3200Å^3. We hypothesise that such a difference in pocket volume changes in the presence of different GEFs suggests that the X-ray structure has perhaps captured the Rac1-GEF complexes either just after GDP dissociation (PREX1 and DOCK2) or in a state that was pre-organised for GTP:Mg^{2+} association (TIAM1 and VAV1). The changes in the pocket volumes for the nucleotide binding site can be attributed to the changes in the switch region dynamics that govern the traffic in the nucleotide site. The P-loop dynamics also govern the pocket changes that are evident from the GEF- and nucleotide-free states, where the P-loop is more flexible (see figure 4C).

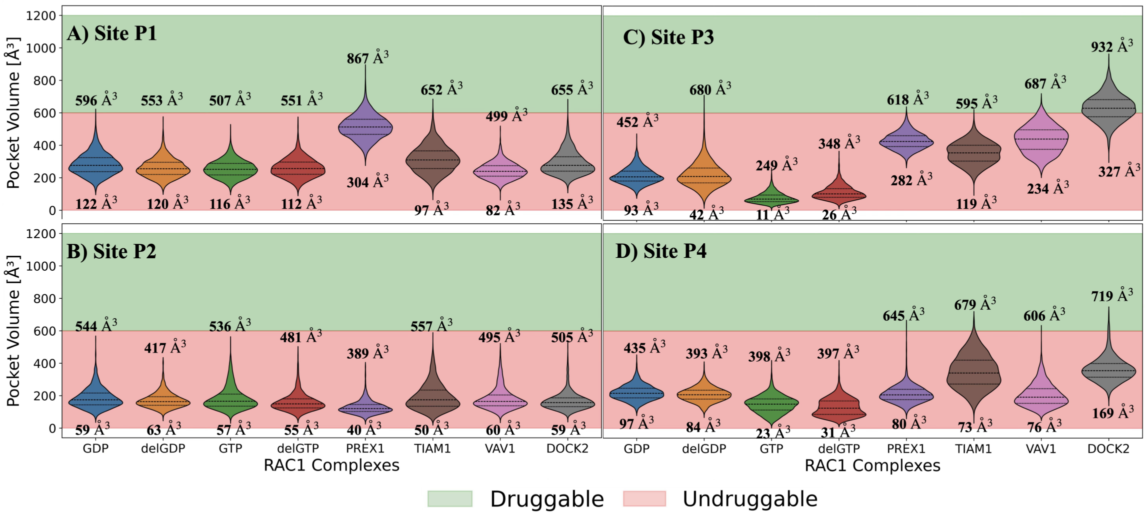

Figure 2: Four binding sites identified using POVME2. **A)** Shows the surface model of Rac1 protein with coloured spears illustrating the binding site identified. The following violin plots illustrate the distribution of the pocket volumes for each of these binding sites on the Rac1 surface monitored when Rac1 is complexed with either GEFs, GDP, GTP or in the *apo*-state. **B)** Site P1. It is the same site of the KRAS-G12C covalent inhibitors bind. **C)**. Site P2 is found at the bottom of the switch I and II loops. **D)**. Site P3 is found at the bottom of the switch I region. **E)**. Site P4 is the only pocket that is not located close to switch I or II regions. The threshold for druggability of a pocket was set according to the following references.[11–13]

We identified four pockets on the Rac1 surface that we believe are potential new binding sites. The Sites P1 and P2 are located in the region lining the switch regions I and II. These sites form the docking site for the GEFs that causes the opening of these sites large enough to attain volumes that can accommodate small molecule binders. In the absence of the GEFs, these sites are too small and hence not detected GEF-free Rac1(**Figure 1A**). Site P1 in Rac1 is similar to the one in K-RAS and H-RAS[14,15] that is occupied by the covalent inhibitors in G12C RAS variants. Sites P1 and P2 are present at a crucial location that could modulate the dynamics of the switch regions. We hypothesise that one of the consequences of targeting Site P1 and P2 is, first, that the presence of a tight-binding molecule could prevent the recruitment of GEFs. Second, the molecules occupying these sites could act as molecular glues that stabilise or facilitate the interaction between Rac1 and GEFs. Both scenarios suggest that the modulation perturbs the GEF-mediated downstream signalling (**Figure2B**). Due to the differences in the binding volumes caused by different GEFs, we anticipate that this can be handled to develop molecules that could target Rac1 in GEF GEF-specific manner. It is also our belief that there exists a possibility to occlude GTP binding, albeit non-competitively, by preventing the dissociation of GEFs that is crucial for nucleotide exchange. Thereby prolonging the nucleotide-free state and preventing the downstream signalling. Site P2 is located below Site P1, and also found to marginal increase in the pocket volume in the presence of GEFs (**Figure 2C**). We propose that, owing to the proximity, this could serve as an extension for molecules binding to Site P1 (**Figure1A**).

Away from the GEF binding site are located sites P3 and P4 (**Figure 1A**). Site P3 is located at the starting point of switch I. Targeting this site could modulate the switch I dynamics, which in turn could have an influence on the traffic to the nucleotide-binding site as well as GEF binding. Residues lining the site P3 are known to interact more specifically with DHR-containing GEFs belonging to the DOCK family (*Add reference for PDB: 7DPA*). The ELMO1 protein that facilitates the binding of the DOCK family of GEFs is observed to interact with residues that form the site P4. Taken together, we hypothesise that modulation of site P3 and P4 by small molecules could specifically target the pathologies arising from abnormalities related to the DOCK family of GEFs.

Accelerating the pocket opening using cosolvents

Transient pockets open in response to ligand binding, suggesting an induced fit mechanism. However, in the absence of known binders, cosolvents that structurally resemble drug-like fragments are used as chemical probes to accelerate the opening of binding sites that require ligand binding. In this work, we used three commonly used cosolvents with different hydrophobicities to further the nature of the binding site. PBD entry with accession number 3TH5 is the only X-ray crystal structure that is crystallised in the active-like conformation bound to GTP analogue, Guanylyl imidodiphosphate (GNP), with any regulatory protein. RAC1 conformation in this case shows that all GEF binding induced pockets are in the undruggable state (small pocket volumes). We performed a microsecond timescale MD simulation to check the effectiveness of the chemical probes to open the transient pockets. We observe the three sites, P1, P3 and P4, visit conformations where the pocket volumes cross the threshold and are open enough to be druggable. However, systems without cosolvent probes remain below the threshold (Figure3).

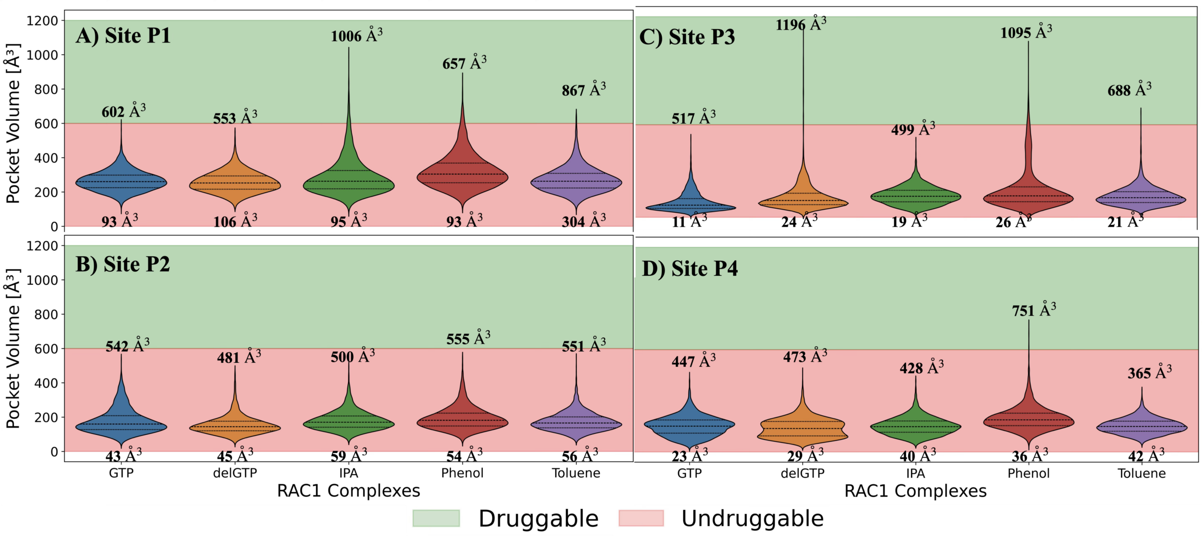

Figure 3: Accelerating transient pocket opening by cosolvent MD simulations.

Role of switch region dynamics in formation of transient pockets.

In RAC1, like all other small GTPases, the switch regions should have much higher conformational movements that are sensitive to nucleotide and GEF binding. The essential dynamics extracted from the dominant modes of motions suggested by the first principal component (XX%) also suggest that switch regions contribute to most of the dynamics in the RAC1. Further, we embarked on understanding how the switch regions' dynamics change in response to GEF binding or nucleotide binding. The nucleotide-free and GEF-free states served to understand the changes inherent to RAC1 without its binding partners.

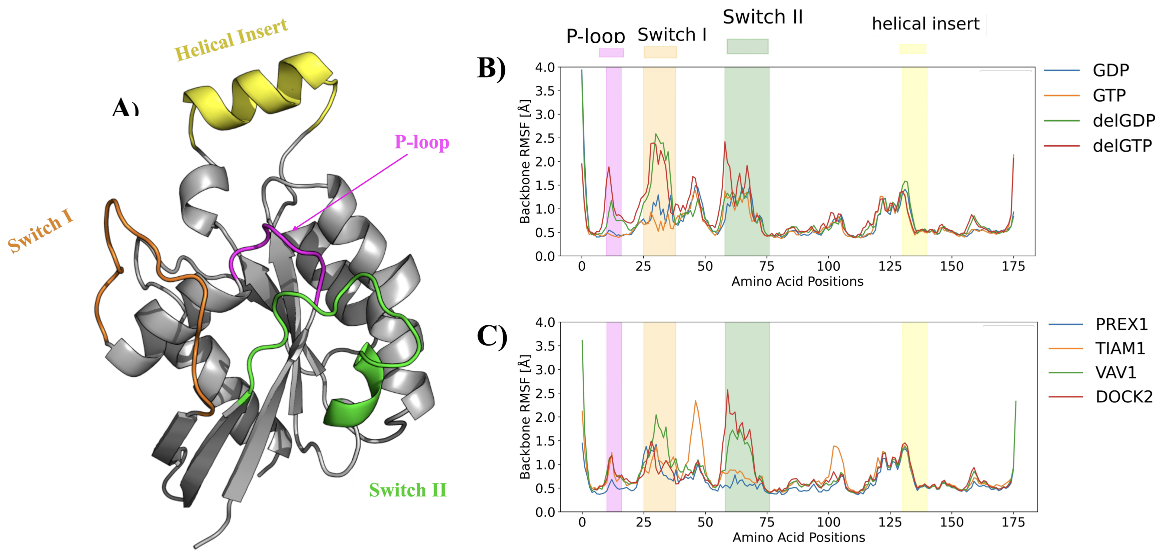

Figure 4: Rac1 domains and dynamics of each region. **A)** A Cartoon representation of human Rac1 (**PDB id:1HE1**). The P-loop region is depicted in the magenta colour. The Switch I region is depicted in Orange Colour, and the Green coloured region is the Switch II region. The helical region is shown in yellow colour. **B)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for the human Rac1 domain to compare the effect of nucleotide binding on the dynamics of various domains. **delGDP** and **delGTP** are systems created by deleting the nucleotides from the Rac1 binding pocket to create corresponding *apo* system. **C)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for the human Rac1 domain to compare the effect of GEF binding on the dynamics of various domains.

Energetic contribution of residues involved in GEF binding

We hypothesise that most of the transient pockets in RAC1 and small GTPases in general open to accommodate the GEFs required for the nucleotide exchange. Therefore, we found it prudent to analyse the energetic contribution of the RAC1 residues that are involved in GEF binding, and their plausible role in binding small molecules. This can

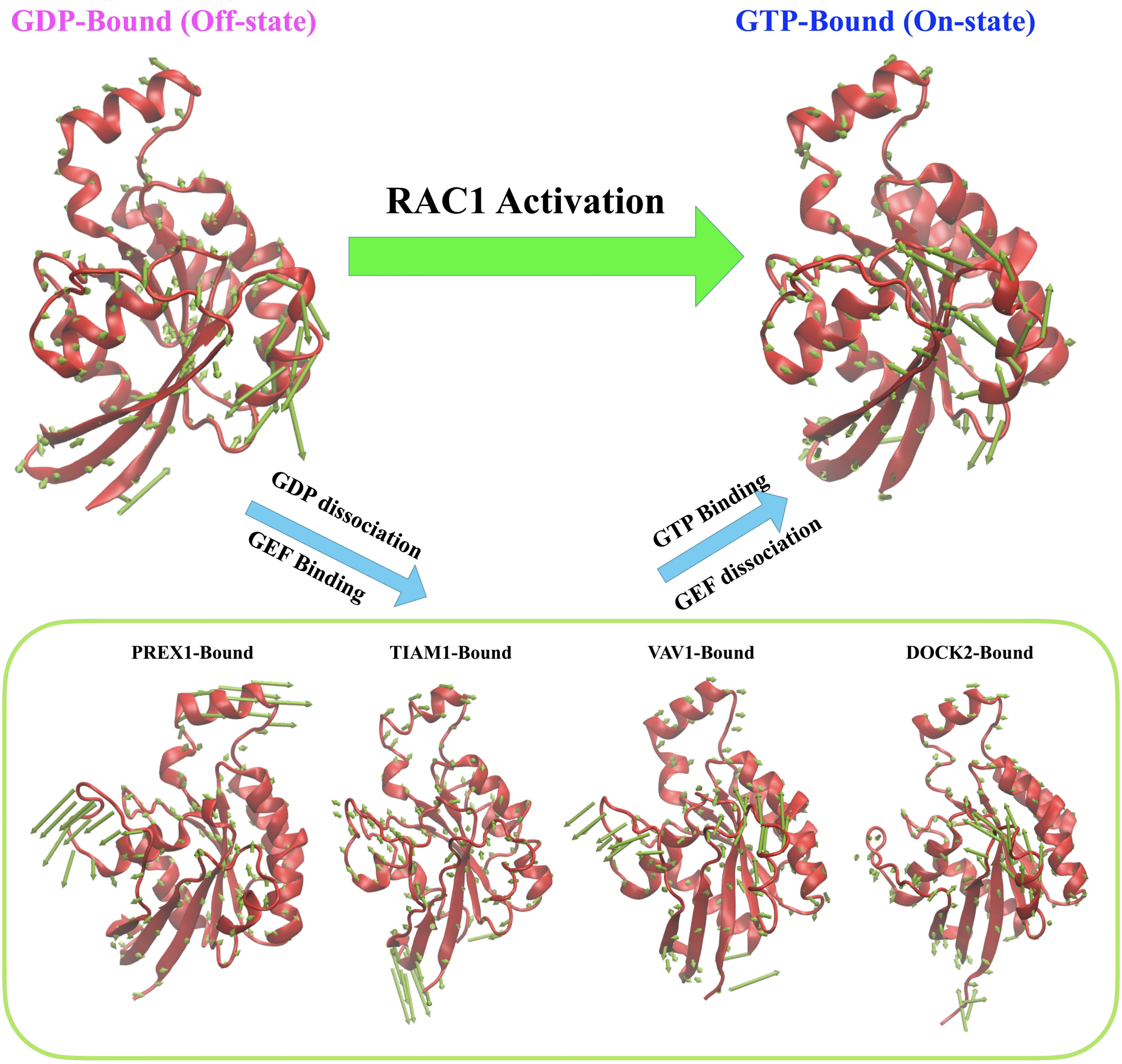

Figure 5: Essential Dynamics of RAC1 proteins to explain the opening and closing of the transient pockets. (TO DO: colour code domains)

enable use to understand the nature of protein-ligand interactions, the small molecules are likely to exhibit. As described earlier, there is a dominant role of the switch regions in GEF binding that leads to dissociation of GDP and association of GTP. This nucleotide exchange, catalysed by the GEF, necessitates their interaction with the switch regions, thus modulating their conformation such that it facilitates this exchange. The sites P1, P2 and P3 are located at the site where GEFs dock to the RAC1 protein to carry out their role. We also observe that the pocket volumes of these sites increase in the GEF-bound state, indicating that these sites are crucial for GEF binding.

Methods

Preparation of systems for molecular dynamics simulation

The coordinates for all complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in **Table 1**. The X-ray structures were imported and processed in the UCSF Chimera v1.17.[16] Any missing loops were built using Modeller[17] with UCSF Chimera v1.17 as the GUI.[18,19]

Table 1: Rac1 conformations and binding partners used in this study

PDB id	Rac1 Modulator/Ligand	Comments
1HE1[20]	GDP:Mg^{2+}	Inactive Rac1
3TH5[21]	GTP:Mg^{2+}	Active Rac1
4YON[22]	PREX1 (aa. 34-366)	DBL family member
1FOE[23]	TIAM1 (aa. 1033-1406, mouse)	DBL family member
2VRW[24]	VAV1 (aa. 170-575, mouse)	DBL family member
2YIN[25]	DOCK2 (aa. 1192-1622)	DOCK family member

Generating AMBER-compatible parameters

All systems were prepared for MD simulations with the Leap module in AMBERtools23. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field.[26] Amber-compatible parameters for all nucleotides were obtained from the Manchester AMBER database. For the VAV1 zinc finger region, the parameters of the Zinc atom, coordinating cysteines and histidine residues, the force field parameters were obtained from the ZAFF[27] force field library.

Molecular dynamics simulations

All complexes were solvated with TIP3P[28] waters enclosed in a box with dimensions extending 10Å beyond any protein atom, and an appropriate number of neutralising ions were added to maintain the system's neutrality. The complexes were relaxed using three cycles of minimisation of 50000 steps each. The first 5000 steps of minimisation adopted the steepest descent algorithm, and the remaining 45000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0Å for the minimisation phase. The first cycle of minimization was performed with 25 kcal/(molÅ^2 restraining force on the solute atoms, this force constant was reduced to 5 kcal/(molÅ^2) in the next cycle, and the final cycle was performed without any restraining forces on either the solute or the solvent atoms. The system was heated in 4 stages (see **Table 2**) from 0 K to 300 K for 10 ns with 50 kcal/(molÅ^2 restraining force on the solute atoms using the Langevin dynamics, with a collisional frequency of 2 ps^{-1}. The next step employed the NPT ensemble for density equilibration for another 2 cycles of 10 ns each, with the restraining force on the solute atoms gradually decreasing from 5 to 0 kcal/(molÅ^2. In the equilibration phase, the cut-off for evaluating non-bonded interactions was placed at 8 Å. In the production phase of the MD simulations, all systems were simulated for 100 ns in triplicate, amounting to a 300 ns trajectory for each system. The long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) summation method.[29] The cut-off for evaluating non-bonded interactions was placed at 9.0 Å for the production phase. All the computations were done using the CUDA-enabled PMEMD code[30–32] in AMBER22.[33,34] All simulations were performed with the SHAKE algorithm[35] to constrain bonds to hydrogen atoms, with the tolerance set to 0.00001 Å. A 2 fs integration time step was used to solve Newton's equations of motion. For analysis, the conformations were saved every 100 ps, amounting to 1000 snapshots per trajectory (3000 per system).

Table 2: Details of stages of Heating using NVT Ensemble

Stage	**Temperature range (K)**	**Time (ns)**
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
Stage 4	300 to 300	7.5 to 10

Measuring the structural dynamics of Rac1

Analysis of the MD trajectory was performed with the cpptraj[36] module in AMBER-Tools23. Before the analysis, all solvent atoms and neutralising ions were stripped off, and the protein conformations were aligned to their respective X-ray conformation. The mass-weighted Cα root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The root-mean-square fluctuations (RMSF) per residue for the backbone and side-chain atoms were computed to understand the domain-specific fluctuations. Furthermore, to understand the specific dynamical changes in Rac1 when bound to different binding partners, we extracted the essential dynamical properties using Principal Component Analysis (PCA). The Variance-Covariance matrix for Cα atoms was extracted in the cpptraj[36] module in AMBERTools23. We extracted and analysed 10 principal components; however, the first two principal components explained $< 50\%$ variance in the data. To visualise the essential dynamics, the vectors were projected on the Cα atoms and visualised in VMD 1.9.3.[37]

Identification and measurement of alternate pockets on Rac1

POVME2[38,39] was employed for identifying various pockets on the Rac1. To be able to incorporate various experimental conformations of Rac1, we employed Rac1 with different binding partners (see **Table 1**). To identify the pockets, the pocketID module on POVME2 was used on the aforementioned Rac1 structures. The parameters used for pocket identification are shown in **Table 3**. Following the pocket identification, we monitored the pocket volume changes of these pockets throughout MD trajectories. It has been observed that the pocket might be a good candidate for druggability if its volume range was found to lie between 610Å^3 to 930Å^3.[11–13]

Table 3: Parameters used for pocket identification

Parameter	**Value**
pocket detection resolution	4.0 Å
pocket measuring resolution	1.0 Å
clashing cutoff	3.5 Å
number of neighbors	4
number of spheres	5
sphere padding	5.0 Å

MD simulations with Cosolvent probes

MD simulations with cosolvent probes are very useful to accelerate pocket opening or to keep the transient pockets open for a longer timescale in MD simulations to better understand pocket characteristics. We identified transient pockets in response to GEF binding, which were otherwise not observed or very hard to locate. To ensure that these transient pockets can remain open sufficiently long to be druggable, we used cosolvent MD simulations. For this purpose, we used 1M concentration of isopropanol, phenol and toluene. The choice of these solvents mimics drug-like fragments with decreasing water solubility. In this case isopropanol (logP = 0.05) > phenol (logP = 1.39) > toluene (logP = 2.56). The simulation box with Rac1 protein (PDB: 3TH5, GTP and Magnesium ions were deleted) and 0.15M of NaCl was prepared using packmol.[40] The protocol for MD simulations is the same as mentioned above. GAFF2 parameters were assigned to the organic solvents. For cosolvent MD simulations, we performed triplicate MD simulations for 1μs, amounting to a total sampling time of 3μs per solvent system. For the baseline calculations, we also prepared two systems without adding cosolvents. One is the RAC1 complexed with GTP and Magnesium ions, which mimics an active-like conformation of RAC1. The apo system was created by deleting GTP and Magnesium ions from the active site system described previously. These systems would enable us to understand if any of the transient pockets can spontaneously open without the influence of the cosolvents and any ligand in the nucleotide pocket.

Quantifying the GEF binding hotspots on Rac1 surface

The free energy of binding was computed using the equation (Eq-1) on the conformations saved from the MD simulations using the MMPBSA module[41] available in AMBER-Tools23;

$$\Delta G_{\text{binding}} = \Delta E_{\text{MM}} + \Delta H_{\text{GBSA}} - T\Delta S \tag{1}$$

Where ΔE_{MM} is the interaction energy between the binding partners computed in the gas phase using a molecular mechanics method. The solvation-free energy or the enthalpy of hydration ΔH_{GBSA} was computed using the Generalised-Born Surface Area (GBSA) method, and the entropic component of binding $T\Delta S$ was not computed due to the huge computation cost. The internal dielectric constant for GBSA calculations was set to 1, and the solvent dielectric constant was fixed at 80. For computing the free energy of binding using the MM-GBSA method, the OBC's GB model[42] ($igb = 5$) was used. The mbondi3 radii set was employed to define the PB radii for all solute atoms during the preparation of the topology and parameter files in the leap module in AMBERTools23. The ionic strength for GBSA calculations was maintained at 150 mM. All energy components, in an implicit solvent, were calculated using a large cut-off of 999 Å. The surface area for the non-polar solvation term was computed using the linear combination of pairwise overlaps (LCPO) method.[43] To identify the critical amino acid residues that are critical for the protein-protein interactions between Rac1 and GEFs, we computed the per-residue decomposition of the binding affinity value. This will also enable us to estimate the contribution from the residues lining the newly identified transient pocket and their energetic contribution to GEF binding.

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